

Oracle Application Software Engineer 1 - What to Learn

Context

This is a SQL/backend-heavy application-support role (Career Level IC1, 0-2+ yrs) on Oracle's healthcare (Cerner Millennium) product line in Bengaluru. It leans hard into SQL, PL/SQL, and application-support/debugging skills rather than the AI/React-native stack your resume otherwise showcases. Below is the delta: what's new, why the JD wants it, and the 20% that matters most — bridged from your existing React/JS/Node background.

MUST – learn before interview

PL/SQL (Oracle's procedural SQL extension)

- What it is: SQL plus procedural constructs (loops, conditionals, variables, stored procedures/functions, triggers, cursors) – Oracle Database's native language for writing business logic that runs inside the DB engine.
- Why the JD wants it: the role explicitly builds/enhances reports via “data fetching through SQL, data processing, and data representation” – PL/SQL is how that logic lives in an Oracle shop.
- The important 20%: `CREATE OR REPLACE PROCEDURE/FUNCTION`, `CURSOR` loops, `%TYPE/%ROWTYPE`, exception handling (`EXCEPTION WHEN . . .`), and basic package structure (spec + body). You already think in functions/parameters from JS – PL/SQL procedures map directly onto that mental model, just with SQL embedded and explicit transaction control (`COMMIT/ROLLBACK`).
- Bridge: think of a PL/SQL procedure like a Node.js service function that takes params, runs queries, and returns/updates rows – except it executes inside the database itself instead of an app server.

SQL for reporting & data analysis (deepen beyond CRUD)

- Why: “develop and design new reports, including data fetching through SQL, data processing, and data representation.”
- The 20%: `JOIN` types (inner/left/right), `GROUP BY` + aggregate functions, window functions (`ROW_NUMBER()`, `RANK()`), CTEs (`WITH` clauses), and query performance basics (indexes, `EXPLAIN PLAN`). Your Skimly project already used raw parameterized Neon/Postgres SQL – this is the same skill, applied to Oracle DB syntax and report-shaped queries instead of app CRUD.

Java (JD lists it as a required programming language)

- Why: “Programming (e.g., Python, Java, C++)” is a required skill – Oracle's healthcare stack (Cerner/Millennium) has a large Java footprint.
- The 20%: core syntax (classes, interfaces, collections – `List/Map`), exception handling (`try/catch`), and how a typical enterprise Java app is structured (layers: controller/service/DAO). You already know C++ and TypeScript – Java's static typing and OOP model is a short hop from both; focus on syntax fluency, not deep frameworks (Spring is unlikely to be probed at IC1 level).

Formal Unit Testing discipline

- Why: explicitly listed as a required skill.
- The 20%: what a unit test is (isolated, repeatable, asserts one behavior), the arrange-act-assert pattern, and naming one framework per language you'd use (JUnit for Java, pytest for Python). None of your shipped projects have real test suites (audit-confirmed) – be honest that you understand the discipline conceptually and are ready to apply it, rather than claiming hands-on production test-writing experience you don't have.

GOOD TO HAVE – lower priority, skim only

Healthcare workflow & the Millennium data model

- Why: this req explicitly says “gain an understanding of healthcare workflows and the Millennium data model” – Millennium is Oracle Health’s (formerly Cerner’s) core EHR platform.
- The 20%: you don’t need deep healthcare-IT knowledge going in – just be able to say you know Oracle Health/Cerner Millennium is a hospital EHR system, and that healthcare data (patient records, orders, results) demands extra care around data integrity and access control. This is explicitly something you’ll be onboarded into, not screened on.

CCL (Cerner Command Language)

- Why: Cerner/Oracle Health’s proprietary report-writing language, layered on top of the Millennium database.
- The 20%: just know it’s a SQL-like scripting language specific to Cerner reporting – don’t invest deep study time pre-interview; it’s an internal tool you’d learn on the job.

BI Tools (Tableau, Business Objects, Power Insight, DA2)

- Why: listed as “good to have” for report/dashboard work.
- The 20%: you already understand dashboards conceptually from building 5 Mixpanel dashboards at Clink AI – if asked, frame that experience as transferable (data -> visual reporting -> stakeholder consumption), and mention you’re comfortable picking up a new BI tool’s UI quickly.

Interview framing tip

This JD doesn’t line up with your AI/React-Native portfolio, and that’s fine – lead with what does transfer: strong SQL fundamentals (Skinly’s raw Postgres SQL), a debugging/RCA habit (Sentry-driven fixes at Clink AI), Git-based release discipline, and fast ramp-up ability (learned Redux, RAG, and EAS/OTA pipelines on the job). Be upfront that Java and PL/SQL are the two genuinely new items you’re picking up for this role.